

# Root Cause Analysis — RCA-2025-005

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## Procedural Shortcut Analysis: CSM-7 Oil-Spectrometer Step Bypass During Peak Demand Windows

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| Field                       | Value                                                                                                                             |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Document ID                 | RCA-2025-005                                                                                                                      |
| Date Issued                 | 2026-02-25                                                                                                                        |
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| Reviewing Manager           | Priya Ramachandran, Plant Manager, NW-PowerPool                                                                                   |
| Approving Engineer          | K. Bhattacharya, Director of Reliability Engineering                                                                              |
| Region                      | NW-PowerPool (with fleet implications)                                                                                            |
| Failure Mode Classification | H-1 (Human-Procedural) + O-1 (Organizational — Fatigue and Pressure)                                                              |
| Distribution                | NW-PowerPool Operations, Reliability Engineering, Compliance, HR / Workforce Operations, Training, Executive Operations Committee |

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### Executive Summary

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Audit of close-out documentation for CSM-7 (Cooling System Maintenance Procedure) work orders at NW-PowerPool in FY2025 found that the **quarterly oil-spectrometer step was bypassed or deferred on 11 of 42 work orders (26%)**. The bypasses cluster around two periods: the Spring Vibration Cascade (RCA-2025-001, May–June) and the run-up to the Fall Overheating Sequence (RCA-2025-002, July–September preparation window).

The bypass pattern is implicated as a contributing factor in at least two major outages:

1. **OIR-2025-002** — WIN-10131 August overheating event. CSM-7 quarterly oil-spectrometer test scheduled for July 2025 (WO-2025-07212) was closed as "deferred to next scheduled outage" per technician notes. Iron particulate at the time of the August trip was 18 ppm (limit 12).

2. **OIR-2025-004** — TRA-10561 December overheating event. The fan-motor resistance check (analogous transformer auxiliary procedure) was omitted from the June 2025 PM (WO-2025-06035), recorded as "deferred to year-end shutdown." Fan bank 3-D failed under load.

This RCA does not assign individual blame. The bypass pattern is a **predictable response to fatigue and pressure** under the FMP-2 violations documented in CAR-2025-001 and to the staffing shortfalls during Spring Vibration Cascade response. The corrective action is **procedural and organizational, not disciplinary**.

## 1. Failure Mode Classification

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**H-1 (Human-Procedural):** Technicians, faced with a multi-hour PM that includes a 90-minute oil-spectrometer step requiring shutdown of the unit and offline lab analysis, defer the step when the schedule is compressed. The procedure permits this in writing — CSM-7 §3.6 allows deferral "with written justification" — but in practice the written justifications are perfunctory or absent.

**O-1 (Organizational):** The factors driving compression are organizational: surge response demands during outage clusters, FMP-2 fatigue cap exceedances, single-depot dispatch constraints (referenced in RCA-2025-001 and RCA-2025-004). The procedure-bypass is a symptom of an over-loaded delivery system.

## 2. Fishbone / Contributing Factors

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### Method:

- CSM-7 §3.6 deferral language is permissive. The procedure permits deferral with "written justification by lead technician." It does not require management approval or backlog tracking.
- There is no formal CSM-7-deferral dashboard. Deferrals are recorded in technician notes and discovered only on audit.

### Manpower:

- FMP-2 cap was exceeded by 4 lead techs during Spring Vibration Cascade (CAR-2025-001). Lead techs were drawn from the same pool that executes quarterly CSM-7 work, creating direct competition between firefighting and PM.
- Cross-training on the oil-spectrometer instrument is limited; 6 of 22 NW-PowerPool technicians are certified. Bench depth is insufficient.

### Method (training):

- The 2024 CSM-7 refresher training included a module on "when to defer" but did not include a module on "deferral cost and downstream risk." The training implicitly normalized deferral.

**Machine:**

– The oil-spectrometer instrument at the Bend depot was out-of-calibration from 2025-03-15 to 2025-05-02. During that window, samples were sent to the regional offsite lab; turnaround increased from 4 hours to 6 days. This significantly extended the time-cost of executing the step and contributed to deferral.

**Materials:**

– Pinnacle Energy Maintenance close-out templates for CSM-7-equivalent work do not include an explicit "oil-spectrometer executed Y/N" field. Pinnacle close-outs at OIR-2025-001, OIR-2025-003, and OIR-2025-009 had missing or ambiguous oil-spectrometer entries. See CAR-2025-004.

### 3. Timeline of Relevant CARs and Outages

| Date                | Event                                                                                            | Reference    |
|---------------------|--------------------------------------------------------------------------------------------------|--------------|
| 2024-12             | CAR-2024-118 opens — preventive maintenance cadence inadequate                                   | CAR-2024-118 |
| 2025-01-13          | WIN-10131 first overheating event of FY25                                                        | OIR-2025-005 |
| 2025-03-15 to 05-02 | Bend depot oil-spectrometer out of calibration                                                   | (depot log)  |
| 2025-04 to 2025-06  | Spring Vibration Cascade peak; multiple CSM-7 deferrals recorded                                 | RCA-2025-001 |
| 2025-07-12          | CSM-7 deferral on WIN-10131 (WO-2025-07212) — directly implicated in 2025-08-28 trip             | OIR-2025-002 |
| 2025-09-23          | Pinnacle close-out on WIN-10011 pitch system (WO-2025-09204) without documented pre-charge check | OIR-2025-009 |
| 2025-10 to 2025-12  | Fall Overheating Sequence; CSM-7 audit initiated 2025-11                                         | RCA-2025-002 |
| 2026-02-25          | This RCA issued                                                                                  | RCA-2025-005 |

### 4. Quantified Business Impact

The bypass pattern is implicated as a contributing factor in two named OIRs (OIR-2025-002 and OIR-2025-004). Attributable lost generation:

| Event                        | Lost MWh | Revenue impact (USD) | Vendor cost (USD) |
|------------------------------|----------|----------------------|-------------------|
| OIR-2025-002 (WIN-10131 Aug) | 4,632.9  | \$277,974            | \$385,000         |
| OIR-2025-004 (TRA-10561 Dec) | 3,558.6  | \$213,516            | \$412,000         |
| Subtotal                     | 8,191.5  | \$491,490            | \$797,000         |
| Total attributable impact    |          |                      | ~\$1.29M          |

The two events together account for approximately 12% of NW-PowerPool's FY25 maintenance cost and 1.2% of stated FY25 net profit (\$103.5M). The cost of executing CSM-7 oil-spectrometer steps on schedule across the affected work orders would have been approximately \$48K of technician time and lab cost.

## 5. Long-Term Corrective Recommendations

1. **Amend CSM-7 §3.6** to require:
2. Management approval (Plant Manager or designee) for any deferral.
3. A specific re-scheduling date within 30 days of deferral.
4. Backlog tracking of all CSM-7 deferrals in the operations dashboard.
5. **Expand technician certification** on oil-spectrometer operation. Target 12 certified technicians at NW-PowerPool by end of Q2 2026.
6. **Recalibrate and add a second oil-spectrometer** at the Bend depot to remove the calibration-window single-point-of-failure. Internal estimate \$42K.
7. **Pinnacle Energy Maintenance close-out template** must include explicit Y/N fields for each CSM-7 sub-step. Refer to CAR-2025-004 and VRA-2025-004.
8. **Address upstream organizational drivers** — FMP-2 enforcement (CAR-2025-001), single-depot footprint (RCA-2025-004 recommendation 5), OEM-bulletin intake SLA (CAR-2025-009). The procedural shortcut is downstream of these.
9. **Training module update.** Replace "when to defer" with "deferral cost and downstream risk" in the 2026 CSM-7 refresher.
10. **No individual discipline.** This RCA explicitly recommends against treating documented deferrals as performance issues. The structural causes are organizational; punishing individual technicians will reduce documentation quality further, which is the opposite of what is needed.

## 6. Vendor Performance Commentary

**Pinnacle Energy Maintenance** is the primary external contractor for CSM-7 work at NW-PowerPool-WIND-003 and -024. Their close-out template inherits from a pre-acquisition predecessor company (acquired 2022) and references deprecated procedure numbers. This documentation drift is independent of the deferral pattern but compounds it — when Pinnacle close-outs are ambiguous, deferral becomes harder to detect. Refer to CAR-2025-004.

**Cascadia Power Services Inc.** internal CSM-7 close-outs are higher quality but not perfect. CPS does record oil-spectrometer results in a structured field; the gap is in deferral justification, which is free-text.

**Heartland Reliability Partners** has not been used at NW-PowerPool for CSM-7 work. The Heartland MSA includes a structured close-out template that could be a model for the Pinnacle redline.

## Signatures

| Role                                | Name                                  | Date       |
|-------------------------------------|---------------------------------------|------------|
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| Operations VP                       | T. Sundgaard                          | 2026-03-05 |

**Distribution:** NW-PowerPool Operations · Reliability Engineering · Compliance · HR / Workforce Operations · Training · Vendor Management · Executive Operations Committee

**References:** OIR-2025-001 · OIR-2025-002 · OIR-2025-003 · OIR-2025-004 · OIR-2025-009 · CAR-2024-118 · CAR-2025-001 · CAR-2025-004 · CAR-2025-006 · CAR-2025-009 · EMR-2025-002 · EMR-2025-005 · RCA-2025-001 · RCA-2025-002 · RCA-2025-003 · RCA-2025-004 · VRA-2025-004 · CSM-7 · FMP-2 · MS-2025-03